Project Title: IkiminaTrack: Digital Savings Group Ledger and Micro-Loan System

Student: Akandwanaho Joshua (20251SEN321)
Course: INSY 8311 & Database Development with PL/SQL
Instructor: Eric Maniraguha

1. Problem Statement

In Rwanda, community savings groups, locally known as Ikimina, are a fundamental financial tool for thousands of citizens, particularly in rural and semi-urban areas. These groups allow members to pool savings and access micro-loans without the strict requirements of traditional banks. However, the vast majority of these groups still rely entirely on paper notebooks and manual calculations to track member contributions, loan disbursements, and interest. This manual approach is highly prone to human error, leads to lost or damaged financial records, and frequently causes disputes among members regarding their balances and late penalties. Furthermore, treasurers lack real-time visibility into the group's total liquidity and default risks, making it difficult to make informed financial decisions.

2. Proposed Solution

The proposed solution is IkiminaTrack, a robust, automated database system built on Oracle Database using PL/SQL. The system will digitize the core operations of a savings group. It will securely store member profiles, track monthly savings contributions, manage micro-loan disbursements, and monitor repayment schedules. By centralizing this data, the system will eliminate manual calculation errors, provide a transparent audit trail, and ensure that the group's financial rules are applied consistently to every member.

3. Data Source

Because actual Ikimina transaction data is highly private, this project will utilize a real, anonymized microfinance dataset from Kaggle (Lending Club Loan Data). This dataset contains thousands of records of loan amounts, interest rates, repayment statuses, and defaults. The data will be adapted to reflect the Rwandan context (e.g., converting loan amounts to Rwandan Francs and adjusting interest rates to match local microfinance standards). It is open-source and licensed for educational use, ensuring compliance with privacy rules.

4. Planned Tables

The database will be normalized to the Third Normal Form (3NF) and will include the following core tables:

· Members: Stores member details (Member_ID, Name, Join_Date, Status).
· Contributions: Records monthly savings (Contribution_ID, Member_ID, Amount, Date).
· Loans: Tracks loans taken by members (Loan_ID, Member_ID, Principal_Amount, Interest_Rate, Status).
· Repayments: Records loan payments (Repayment_ID, Loan_ID, Amount_Paid, Date).

· Audit_Log: Tracks changes to sensitive data (e.g., changing a member's status or overriding a late penalty).

5. Planned PL/SQL Features

· Triggers: To enforce strict business rules. For example, a trigger will prevent a member from taking a new loan if they have an unpaid penalty or an active defaulted loan. Another trigger will automatically log any unauthorized attempts to delete a financial record.
· Stored Procedures: A procedure named record_contribution() will accept a payment and automatically split it between the member's savings and their outstanding loan interest.
· Functions: A function calculate_outstanding_balance(member_id) will return the total amount a member owes (principal + accrued interest).
· Cursors: A cursor will be used at the end of every month to loop through all active loans and calculate/apply the monthly accrued interest.
· Packages: A package pkg_financials will group related procedures and functions for the savings and loan logic, improving database performance and organization.

6. Innovation

The innovation for this project will be an Analytics Dashboard built using Microsoft Power BI, connected directly to the Oracle Database. This interactive dashboard will allow the group treasurer (who may not know SQL) to monitor the group's financial health. The dashboard will visually highlight members with overdue loans, display the group's total available capital, and track monthly contribution trends. Note: All business rules and restrictions will remain strictly enforced by PL/SQL on the database side; Power BI will only serve as the visualization tool.

7. References (APA)

FinScope Rwanda. (2020). Financial Inclusion in Rwanda. Retrieved from https://www.finscope.org/

Lending Club. (2023). Loan Data. Kaggle. Retrieved from https://www.kaggle.com/datasets/wordsforthewise/lending-club

Mwangi, P., & Karanja, J. (2019). The role of digital systems in informal savings groups. Journal of African Finance, 12(3), 45-60.

National Bank of Rwanda. (2022). Annual Financial Stability Report. Retrieved from https://www.bnr.rw/

Oracle Corporation. (2023). Database PL/SQL Language Reference. Retrieved from https://docs.oracle.com/en/database/